frequently a poor interpretation of what is actually observed." Francis specialized in the physics of fluids, and was suspicious of the mathematically simple yet difficult-to-visualize quantum theory. "There were very strong similarities between the mathematics of fluid flow and the mathematics of quantum mechanics, and [my dad] didn't understand quantum mechanics," the younger Clauser recalled. "And he kind of pre-programmed me as the guy who might help try to solve the problem that he couldn't solve." When it came time for John to go to Caltech, he learned quantum physics at the feet of Richard Feynman himself—but he could never shake the suspicion that something was wrong with the theory. His doubts followed him to his PhD program at Columbia, where he learned more about the debates that had hounded quantum physics since its inception. "I… was struggling, trying to understand quantum mechanics. I had read EPR's paper, and also had read Bohm's and de Broglie's work. While I had difficulty understanding the Copenhagen interpretation, the arguments by its critics seemed far more reasonable to me at that time," Clauser recalled. "I also found EPR's arguments much more persuasive than Bohr's.… Hidden variables thus seemed (to me then) to be a perfectly logical solution to the problem. By holding that opinion… I was certainly branded as a heretic by many, and undoubtedly as a quack by others."

Given his background, the title of Bell's paper immediately captured Clauser's attention—and the elegant proof contained within the short paper came as a massive shock. "I said, 'this can't possibly be true,'" Clauser recalled later. "I [thought] it should be easy to find a counter-example. I worked and worked and worked.… I can't find a counter-example. Well, Bell's got to be wrong on the proof. But nope, can't find anything wrong with it. Bounced back and forth between those two [ideas] and finally it just hit me:Jesus Christ, this is a very important result." Clauser, an experimental physicist down to his bones, immediately wondered: Could Bell's idea be put to the test?

Clauser knew there was a chance that Bell's theorem had already been tested inadvertently, as part of some earlier experiment. And, even if it hadn't been, Clauser needed to search the relevant experimental literature, to find out how such an experiment could best be done. He already knew that Chien-Shiung Wu, a professor at Columbia renowned for her work in nuclear physics, had done an experiment similar to the EPR thought